Fixed-Edit FIBER–GRAND: Mandatory Pre-Manuscript Baseline and Tail Closure

Independent audit of commit `2f67a32` and a frozen final computational gate

Scientific de-risking dossier, version 5.0

6 August 2026

Authoritative status

Commit `2f67a32` validly passed its frozen version-4 contract and provides strong evidence for a narrow fixed-edit theory/algorithm paper. It does not yet justify freezing a flagship IEEE Transactions on Information Theory manuscript because the strongest feasible syndrome-trellis reference was disabled at several larger operating points and the reported high quantiles were based on only twelve blocks per largest-length configuration. The correct action is therefore

`CONTINUE_TO_ONE_MANDATORY_PRE_MANUSCRIPT_CLOSURE_GATE`.

A positive version-5 outcome authorizes the manuscript and external review. A negative outcome narrows the work to a rigorous fixed-edit theorem/algorithm paper and stops further performance escalation.

Contents

1 Purpose and scientific decision

The narrowed FIBER–GRAND programme considers exactly t uniformly located deletions followed by independent BSC substitutions on the $m = n - t$ surviving symbols. It generates inverse histories in probability order, canonicalizes repeated candidates, calls only a code-membership oracle at its weakest interface, exactly scores discovered codewords, and terminates when a strict frontier bound excludes every undiscovered codeword and every unseen ML tie.

The version-4 WSL reproduction established correctness and a large compiled timing signal. The machine verdict was `PROCEED_TO_FLAGSHIP_TIT_MANUSCRIPT_AND_EXTERNAL_REVIEW`. Independent inspection agrees that a quality IEEE TIT paper is plausible, but finds two unresolved defects in the *performance evidence*, not in the core mathematics:

(i) the syndrome-trellis exact decoder was disabled at $n = 48$ and $n = 64$, including several points where its declared dynamic-programming budget made it feasible; and

(ii) $p95$ and $p99$ summaries at $n = 64$ were computed from only twelve trials, while the low-p channel makes the zero-substitution event dominant.

Version 5 is the smallest experiment that resolves these two issues without reopening the already closed broad proposal.

1.1 Three outcomes

`AUTHORIZE_FULL_TIT_MANUSCRIPT_AND_EXTERNAL_REVIEW`
The exact theory remains intact, feasible trellis references are enabled, the advantage survives at supported $p99$ and on positive-substitution strata, and manuscript construction is authorized.

`NARROW_TO_FIXED_EDIT_TIT_THEORY_ALGORITHM_PAPER`
The mathematics survives but the flagship performance claim does not. Prepare a rigorous theory/algorithm paper, retain the negative boundary, and stop performance escalation.

`STOP_FLAGSHIP_PERFORMANCE_CLAIM`
A correctness or theorem audit fails. Only a demonstrable implementation repair is allowed.

2 What commit `2f67a32` established

2.1 Exactness and certificates

The reproduced audit reported zero failures in 50,688 likelihood-recurrence cases, 200 complete-tie one-deletion cases, 105 complete-tie two-deletion cases, 15,174 VT checks, 1,602 shell-certificate cases, and 36 moment-sandwich checks. The maximum likelihood-recurrence error was approximately 2.2×10^{-16}. The C++ self-test additionally checked 600 complete small-code worlds.

These tests support the following hard-decision result. Let η_J be the probability of the next unprocessed history in deletion-set stream J. If $s^\star$ is the exact score of the best discovered codeword and

$$s^\star > U_{\text{new}} := \sum_{J \in \mathcal{J}_t} \eta_J,$$

then the discovered codewords at score $s^\star$ form the complete ML tie set. Strictness is essential: equality cannot certify uniqueness or tie completeness.

2.2 Candidate-volume and oracle-relative theory

For observation y, define the best-alignment mismatch

$$\delta_y(x) = \min_{J \in \mathcal{J}_t} d_J(x, y)$$

and the inverse candidate volume

$$V_y(w) = \left|\{x \in \{0,1\}^n : \delta_y(x) \le w\}\right|.$$

With

$$B_m(w) = \sum_{i=0}^{\min\{w,m\}} \binom{m}{i},$$

the audited bounds are

$$2^t B_m(w) \le V_y(w) \le 2^t \binom{n}{t} B_m(w). \tag{1}$$

Every complete candidate has exactly $\binom{n}{t}$ compatible inverse histories, one for every deletion subset.

For $p < 1/2$, set

$$a = \frac{1-p}{p}, \qquad L_{n,t} = \left\lfloor \log_a \binom{n}{t} \right\rfloor.$$

If an ML codeword has best-alignment mismatch $d^\star$, then an exact L0 membership-only decoder must query every candidate in shell $d^\star - L_{n,t} - 1$ that is strictly more likely, unless it has already identified the entire code. FIBER certifies by shell $d^\star + L_{n,t}$. This yields the oracle-relative sandwich

$$V_y(d^\star - L_{n,t} - 1) \le Q^\star_{\mathrm{L0}} \le Q_{\mathrm{FIBER}} \le V_y(d^\star + L_{n,t}), \tag{2}$$

under the original arbitrary-code oracle model.

2.3 Known-cardinality refinement

The code cardinality M is often known. This slightly weakens a pure indistinguishability converse because an unqueried high-likelihood candidate can be excluded after all M codewords have already been found. The correct refinement is the following.

Proposition 1 (Known-cardinality L0 lower bound)**.** *Let*

$$H_y(c^\star) = \left|\{x : \Lambda_y(x) > \Lambda_y(c^\star)\}\right|.$$

An exact decoder that knows only the membership oracle and code cardinality M must, before certifying $c^\star$, either query every member of this higher-likelihood set or have already received M positive membership responses. Hence its total query count satisfies

$$Q^\star_{\mathrm{L0},M} \ge \min\{H_y(c^\star), M\},$$

with the additional trivial requirement of at least one positive query when a codeword is returned.

Proof. Assume some higher-likelihood candidate remains unqueried and fewer than M codewords have been found. The query transcript is then consistent with another size-M code whose membership set contains the unqueried candidate and preserves all previous answers. Returning $c^\star$ would be incorrect for that oracle. Therefore one of the two stated conditions is necessary. □

For a rate-R code, $M = 2^{nR}$. When the interior-shell exponent $h_2(q)$ is below R, the cardinality truncation does not alter the $h_2(q)$ lower exponent. This closes an important assumption gap while preserving the intended favorable-regime theorem.

2.4 Moment, tail, and phase laws

For fixed t, the probability-ordered rank R_{hist} at which the actual inverse history is revealed satisfies, for every $\rho > 0$,

$$\lim_{n\to\infty} \frac{1}{n} \log_2 \mathbb{E}[R_{\text{hist}}^{\rho}] = \rho H_{1/(1+\rho)}(p).$$

Certified FIBER may stop earlier, so only the corresponding upper moment and tail bounds hold universally for Q_{FIBER}. The typical fixed-quantile exponent is $h_2(p)$, whereas the mean-history exponent is $H_{1/2}(p)$. A parallel hybrid with a code-side exact search of exponent $1 - R$ therefore has distinct predicted boundaries

$$h_2(p_{\text{typ}}) = 1 - R, \qquad H_{1/2}(p_{\text{mean}}) = 1 - R.$$

At $R = 0.75$, for example, $p_{\text{typ}} \approx 0.04169$ but $p_{\text{mean}} \approx 0.00903$. A decoder can thus win in median latency while losing in mean or rare-tail work.

3 Why the version-4 performance gate is not yet final

3.1 Feasible syndrome-trellis points were disabled

The compiled syndrome reference estimates its preprocessing work as

$$U_{\text{tr}}(n, r) = 2n^2 2^r, \qquad r = n - k.$$

The version-4 profile supplied a nonzero trellis budget at $n = 32$ but left the global budget at zero for the $n = 48$ and $n = 64$ schedules. Consequently the result files record a trellis availability fraction of one at $n = 32$ and zero at $n = 48, 64$. Yet several larger points are computationally feasible:

Table 1: Charged trellis update estimates.

n	R	r	$U_{\text{tr}}(n, r)$	Version-5 treatment
48	0.75	12	18,874,368	mandatory
48	0.875	6	294,912	mandatory
64	0.875	8	2,097,152	mandatory
64	0.75	16	536,870,912	recorded but not mandatory

At the largest lengths the spectacular version-4 ratios were therefore primarily FIBER versus prefix A*, not FIBER versus the faster of every feasible implemented exact decoder. This does not invalidate the signal; it prevents treating it as closed.

3.2 Twelve samples cannot support a $p99$ claim

The version-4 standard profile used 24 trials at $n = 32$, 16 at $n = 48$, and only 12 at $n = 64$. An empirical $p99$ from 12 observations is essentially an interpolation near the maximum, not a stable high-quantile estimate. Bootstrap confidence intervals on the median do not repair this tail limitation.

The low-p regime also makes zero-substitution blocks dominant. With $m = 63$ survivors,

$$\Pr\{E = 0\} = (1 - p)^{63},$$

which equals approximately 0.8815, 0.7292, and 0.5309 for $p = 0.002$, 0.005, and 0.01, respectively. A large unconditional speedup could therefore describe the easiest $E = 0$ case while hiding a failure on the positive-error blocks responsible for the tail.

3.3 Baseline switching contaminates slope interpretation

Version 4 reported favorable slopes from $n = 32$ to 64, but its authoritative baseline changed from the minimum of trellis and prefix at $n = 32$ to prefix alone at $n = 48, 64$. A slope comparison across changing reference classes cannot by itself establish scaling superiority. Version 5 keeps the trellis enabled at every required feasible point and records which baseline wins each block.

4 Version-5 computational protocol

4.1 Implementations and fairness controls

The package compiles FIBER, prefix A*, and syndrome-trellis search from the same C++20 translation unit using

`-O3 -DNDEBUG -std=c++20 -march=native`.

It retains the exact 600-case C++ self-test and all Python theorem/exactness audits. Each configuration receives warmup blocks excluded from timing; decoder order is randomized; each reported block time is the median of three repeats; and thread-related numerical-library environment variables are fixed to one.

The authoritative per-block reference is the faster certified result among prefix A* and the syndrome trellis whenever the latter is feasible. Preprocessing is included in each decoder time.

4.2 Frozen experiment matrix

The standard profile uses:

- $n \in \{32, 48, 64\}$, $R \in \{0.75, 0.875\}$;
- $p \in \{0.002, 0.005, 0.01\}$;
- random systematic linear and reproducible CRC-polynomial linear codes;
- 120 trials at $n = 32$, 160 at $n = 48$, and 600 at $n = 64$ per family/rate/p configuration;
- three timing repeats and three warmup blocks per configuration;
- exact syndrome membership without a codebook table.

The $n = 64$ schedule yields about 71 positive-error samples even at $p = 0.002$ in expectation, enough to examine the $E = 1$ stratum rather than infer everything from $E = 0$.

4.3 Tail and conditioning outputs

The package reports:

- median ratio with a 95% bootstrap interval;
- empirical $p90$, $p95$, $p99$, maximum, and an order-statistic interval for $p99$;
- separate summaries for $E = 0$, $E = 1$, $E = 2$, and $E \geq 3$;
- median and high-quantile history counts;
- trellis availability, certification, update estimate, and fraction of blocks on which it is the faster reference;
- scaling slopes using a consistent enabled-reference policy.

5 Frozen decision contract

Let a *key configuration* be a largest-length point with $p \leq 0.01$ inside the predicted typical-favorable region. Continuation requires all of the following:

G1. all exactness, candidate-volume, ambiguity, known-cardinality, moment/tail, and phase audits pass;

G2. at least ten key configurations are present and each has at least 500 trials;

G3. FIBER wins the median by at least $1.25\times$ in at least eight key configurations;

G4. the upper 95% bootstrap endpoint of the median ratio is at most 0.8 in at least six configurations;

G5. the empirical $p99$ ratio is at most one in at least six configurations, and the upper order-statistic interval endpoint is at most 1.25 in at least four;

G6. every predicted mean-favorable largest-length configuration has $p99$ ratio at most one;

G7. at least four favorable slope comparisons with a consistent reference-availability policy have disadvantage no greater than 0.08 in log-two wall-time slope;

G8. at least four positive-error strata have at least 50 blocks, median ratio at most 0.8, and $p95$ ratio at most 1.25;

G9. the syndrome trellis is available and certified for all required $n = 48$ points and for $n = 64, R = 0.875$;

G10. the true linear-time VT specialization boundary remains exact.

These thresholds are demanding by design. The purpose is not to rescue the performance claim; it is to decide whether the claim is strong enough to carry a flagship manuscript.

6 Novelty boundary and publication interpretation

The following are established or substantially anticipated: additive GRAND, candidate-likelihood guessing for general DMCs, priority-first decoding, synchronization sequential decoding, hidden-alignment summation, best-string versus best-path distinctions, fixed-k deletion likelihoods, and specialized VT/GC/polar synchronization constructions [1, 2, 3, 4, 5, 6].

The residual contribution is narrower:

> An exact membership-first aggregate inverse decoder with a strict complete-tie certificate for fixed edits, joined to candidate-volume bounds, an L0 oracle-query converse and polynomial competitiveness theorem, fixed-edit moment/tail laws, and a rate-complexity phase diagram.

The present search did not locate this complete package in one primary source. The counting bounds are elementary, and the oracle converse uses a standard indistinguishability technique; their value comes from the integrated theorem-and-decoder result rather than from claiming a wholly new mathematical primitive.

A positive version-5 result supports a high-quality IEEE TIT manuscript. It does not establish field-defining status. That term remains retrospective and would require broader independent uptake and demonstrated relevance beyond one fixed-edit channel.

7 Exact next step after the run

If version 5 returns `AUTHORIZE_FULL_TIT_MANUSCRIPT_AND_EXTERNAL_REVIEW`, the next deliverable is one full IEEE TIT manuscript and a minimal standalone external-review package containing:

1. the manuscript PDF and source;
2. a compact proof supplement only if page limits require it;
3. the claim-by-claim novelty matrix;
4. the exactness and compiled closure evidence;
5. a reviewer handoff requesting one of four classifications: flagship candidate, strong non-flagship paper, blocking mathematical defect, or blocking novelty defect.

If version 5 returns `NARROW_TO_FIXED_EDIT_TIT_THEORY_ALGORITHM_PAPER`, the theorem and exact decoder should still be written, but all flagship timing and field-defining language should be removed. If it returns `STOP_FLAGSHIP_PERFORMANCE_CLAIM`, no further performance campaign is justified.

8 Conclusions

Commit `2f67a32` is a valid and encouraging result. It establishes real flagship-paper potential but not manuscript readiness. The remaining uncertainty is sharply localized: whether FIBER's compiled advantage survives the strongest feasible exact reference and the positive-substitution tail. Version 5 closes that uncertainty without broadening the theory, changing the decoder, or moving the decision thresholds after observing the outcome.

References

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